APPENDIX A
PROOF OF THEOREM 1

Theorem 1. *Assume that node representations are biased and can be discriminated by the predictor, i.e., $\mu_c > \bar{\mu}_c$, and that the hypergraph follows the homophily assumption, i.e., $P_i^{homo} > P_i^{hete}$. Considering the hypergraph message-passing process defined as $x_i^* = x_i + m_{\mathcal{R}(v_i)}$, we have:*

$$\mathbb{E}\{D_\phi(s|x_i^*) - D_\phi(\bar{s}|x_i^*)\} > \mathbb{E}\{D_\phi(s|x_i) - D_\phi(\bar{s}|x_i)\}, \quad (23)$$

which means updated node representations would cause more sensitive information leakage.

Proof. Since D_ϕ is a linear function, we have

$$\begin{aligned}
&\mathbb{E}\{D_\phi(s|x_i^*) - D_\phi(\bar{s}|x_i^*)\} \\
=&\mathbb{E}\{D_\phi(s|x_i) + D_\phi(s|m_{\mathcal{R}(v_i)}) - D_\phi(\bar{s}|x_i) - D_\phi(\bar{s}|m_{\mathcal{R}(v_i)})\} \\
=&\mathbb{E}\{D_\phi(s|x_i) - D_\phi(\bar{s}|x_i)\} + \mathbb{E}\{D_\phi(s|m_{\mathcal{R}(v_i)}) - D_\phi(\bar{s}|m_{\mathcal{R}(v_i)})\}
\end{aligned}$$

For the term $\mathbb{E}\{D_\phi(s|m_{\mathcal{R}(v_i)}) - D_\phi(\bar{s}|m_{\mathcal{R}(v_i)})\}$, we have:

$$\begin{aligned}
&\mathbb{E}\{D_\phi(s|m_{\mathcal{R}(v_i)}) - D_\phi(\bar{s}|m_{\mathcal{R}(v_i)})\} \\
=&\sum_{e_j \in \mathcal{R}(v_i)} \Big[\frac{P_i^{homo}}{d(v_i)}\mathbb{E}\{D_\phi(s|\tilde{x}_{i,j}^{homo}) - D_\phi(\bar{s}|\tilde{x}_{i,j}^{homo})\} - \\
&\frac{P_i^{hete}}{d(v_i)}\mathbb{E}\{D_\phi(s|\tilde{x}_{i,j}^{hete}) - D_\phi(\bar{s}|\tilde{x}_{i,j}^{hete})\}\Big] \\
=&\frac{1}{d(v_i)}\sum_{e_j \in \mathcal{R}(v_i)} (P_i^{homo} - P_i^{hete})(\mu_c - \bar{\mu}_c) \\
=&(P_i^{homo} - P_i^{hete})(\mu_c - \bar{\mu}_c) > 0
\end{aligned}$$

Thus, we can obtain that:

$$\begin{aligned}
&\mathbb{E}\{D_\phi(s|x_i^*) - D_\phi(\bar{s}|x_i^*)\} \\
=&\mathbb{E}\{D_\phi(s|x_i) - D_\phi(\bar{s}|x_i)\} + (P_i^{homo} - P_i^{hete})(\mu_c - \bar{\mu}_c) \\
>&\mathbb{E}\{D_\phi(s|x_i) - D_\phi(\bar{s}|x_i)\}
\end{aligned}$$

□

APPENDIX B
OVERALL TRAINING ALGORITHM OF FAIRHSSL

To generate fair and informative hypergraph representations, we incorporate the proposed two-level augmentation strategy into SE-HSSL, obtaining its fairness-aware variant, FairHSSL. The detailed training algorithm of FairHSSL is illustrated in Algorithm 1. In practice, we find that generating a fair augmented structure with the proposed per-hyperedge sampling scheme is computationally expensive, making it less suitable for large-scale hypergraphs. To address this, in the preprocessing stage, we first conduct orthogonal projection on $\mathbf{X}$ to get the debiased features $\mathbf{X}'$ (lines 2-4) and then perform over-sampling for each hyperedge only once to obtain the rebalanced structure $\mathbf{H}'$ (lines 6-16). Given the fair hypergraph $\mathcal{H}'(\mathbf{X}', \mathbf{H}')$ and the original biased data $\mathcal{H}(\mathbf{X}, \mathbf{H})$, we apply the feature masking and membership masking techniques described in Section III-B to generate fair and biased augmented views efficiently in each training iteration (lines 20-21). Finally, by maximizing the agreement between the biased and the de-biased augmented view with three sampling-efficient signals proposed in Section III-D (lines 23-28), we can eliminate the effect of sensitive information from the encoder during training.

Algorithm 1 Training Algorithm of FairHSSL

Input: $\mathcal{H}(\mathbf{X}, \mathbf{H})$: the original hypergraph; $\mathcal{V}, \mathcal{E}$: node set and hyperedge set; T: epoch number; β: rebalance coefficient; $\lambda_1, \lambda_2, \lambda_3$: hyper-parameters for different loss terms;
Output: Trained hypergraph encoder f_θ

1: `// Feature-level debiasing`
2: Compute sensitive group direction d_{g_0}, d_{g_1} as Eq. (17)
3: Compute bias direction d_{bias} as Eq. (18)
4: Orthogonally Projecting $\mathbf{X}$ into $\mathbf{X}'$ as Eq. (19)
5: `// Structure-level debiasing`
6: **for** $j = 1, ..., |\mathcal{E}|$ **do**
7: **if** $g_{minor}^{(j)} \neq g_{major}^{(j)}$ **then**
8: Compute $N_{os}^{(j)} = \lceil \beta(N_{major}^{(j)} - N_{minor}^{(j)}) \rceil$
9: **if** $N_{minor}^{(j)} = 0$ **then**
10: over-sample $N_{os}^{(j)}$ nodes in $g_{minor}^{(j)}$ from $\mathcal{V}$
11: **else**
12: over-sample $N_{os}^{(j)}$ nodes in $g_{minor}^{(j)}$ from e_j
13: **end if**
14: **end if**
15: **end for**
16: Get rebalanced structure $\mathbf{H}'$ and debiased data $\mathcal{H}'(\mathbf{X}', \mathbf{H}')$
17: `// Self-supervised training`
18: **for** $t = 0, ..., T-1$ **do**
19: `// Epoch-level augmentation`
20: Generate fair view $\tilde{\mathcal{H}'}$ from $\mathcal{H}'$ by Eq. (6-7)
21: Generate biased view $\tilde{\mathcal{H}}$ from $\mathcal{H}$ by Eq. (6-7)
22: `// Sampling-efficient signals`
23: Compute node-level loss $\mathcal{L}_N$ according to Eq. (10)
24: Compute group-level loss $\mathcal{L}_G$ according to Eq. (11)
25: Compute membership-level loss $\mathcal{L}_{HM}$ by Eq. (13)
26: `// Joint optimization`
27: $\mathcal{L} = \mathcal{L}_N + \lambda_1\mathcal{L}_G + \lambda_2\mathcal{L}_{HM} + \lambda_3||\mathbf{\Theta}||^2$
28: Back-propagation to update parameters $\mathbf{\Theta}$
29: **end for**

TABLE A1: Statistics of fairness-sensitive datasets.

Datasets	German Credit	Bail	Credit Defaulter
$\|\mathcal{V}\|$	1000	18876	30000
$\|\mathcal{E}\|$	1000	18876	30000
F	27	18	13
Sensitive Attribute	Gender	Race	Age
Node Labels	Credit status	Bail decision	Future default

APPENDIX C
DATASET DESCRIPTION ON FAIRNESS-SENSITIVE HYPERGRAPHS

The dataset statistics for the three fairness-sensitive datasets are provided in Table A1. A brief overview of these datasets is given below.

- German Credit: The nodes in the dataset represent clients in German Bank, and hyperedges are constructed by linking individuals with the most similar credit accounts to each person in the dataset. The task is to classify credit

TABLE A2: Evaluation results for node clustering: NMI and ARI.

Method	Cora		Citeseer		Pubmed		Cora-CA		NTU2012		Zoo		ModelNet40	
	NMI ↑	ARI ↑	NMI ↑	ARI ↑	NMI ↑	ARI ↑	NMI ↑	ARI ↑	NMI ↑	ARI ↑	NMI ↑	ARI ↑	NMI ↑	ARI ↑
BGRL	40.81	22.54	31.69	24.87	15.98	16.80	32.48	21.72	67.76	33.52	70.55	58.65	73.63	45.36
CCA-SSG	49.08	40.57	38.45	36.02	17.22	17.24	38.13	26.18	73.72	42.26	79.59	74.70	79.08	56.44
COSTA	45.58	36.08	34.83	28.18	20.41	15.92	36.09	22.90	71.77	40.32	76.25	61.95	75.23	46.33
HyperGCL	40.24	32.35	39.06	38.88	28.72	24.73	46.15	39.33	**84.27**	**70.88**	77.40	79.36	93.71	89.91
TriCL	56.95	49.77	42.63	42.44	30.80	29.06	44.39	37.21	83.75	68.33	90.87	88.62	93.80	87.07
HypeBoy	49.33	41.55	38.33	36.05	30.91	27.40	42.58	35.71	77.34	55.69	81.35	73.42	83.99	67.41
SE-HSSL	**59.41**	**53.37**	**43.66**	**43.01**	**32.79**	**29.73**	**50.56**	**44.47**	83.31	68.26	**91.71**	**91.48**	**94.36**	**91.63**

risk levels as high or low based on the sensitive attribute "gender" (Male/Female).

- Bail: The nodes in the datasets are defendants who got released on bail at the U.S state courts during 1990-2009. Hyperedges are constructed based on the similarity of past criminal records among individuals. The task is to classify whether defendants are on bail or not with the sensitive attribute "race" (White/Black).
- Credit Defaulter: The nodes in the dataset represent credit card users, and hyperedges are formed based on the similarity of users' spending and payment patterns. The task is to classify the default status with the sensitive attribute "age" (<25/>25).

APPENDIX D
MORE EXPERIMENT RESULTS

D.1. Node Clustering Task

In this subsection, we conduct the node clustering task on the 7 benchmark datasets. We compare SE-HSSL with the 3 latest HSSL methods and 3 strong graph SSL models. Table A2 shows that SE-HSSL achieves the best clustering results on 6 out of 7 datasets. Similar to the classification task, the generative-based HypeBoy is generally less competitive than SE-HSSL and the other two HCL methods. Furthermore, we find our SE-HSSL achieves an average improvement of 1.80% in NMI and 2.78% in ARI compared to TriCL across all datasets. This is because our sampling-efficient signals effectively reduce training bias and avoid degenerated solutions, making learned embeddings more informative and discriminative.

D.2. Additional Parameter Sensitivity Analysis

Embedding size D. Figure A1(a) illustrates that as we vary the embedding size from 128 to 512, the model performance improves overall, suggesting that SE-HSSL benefits from a larger representation capacity. However, we notice a decline in performance as the embedding size further increases on most datasets. This is because the embedding space becomes sparse when D is too large. Consequently, there is limited meaningful information available in each dimension of the embedding, with many dimensions being irrelevant or redundant. This makes it challenging for the downstream classifier to extract meaningful features from embeddings.

Hyperedge hop K. In the hierarchical membership contrast, we sample neighborhood hyperedges ranging from 1 to K hops for each node. As depicted in Figure A1(b), the model can achieve near-optimal performance when K is set to 1. Increasing the neighborhood range only leads to marginal performance gains on most datasets. This suggests that our model exhibits robustness and is not highly sensitive to the number of hops when sampling hyperedges. We attribute this to our multi-positive membership loss, which does not enforce higher similarity for all positive pairs and effectively disregards noisy positives.

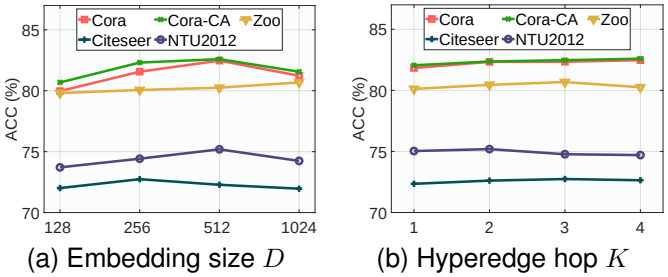
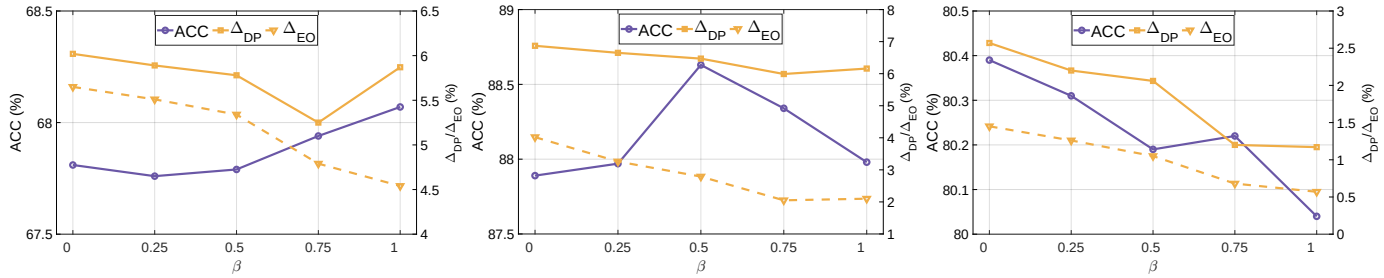


Fig. A1: Parameter analysis on the embedding dimension D and the hyperedge hop number K.

Rebalance coefficient β. We investigate the effect of β within the range [0,1] with an interval of 0.25. As shown in Figure A2, increasing the degree of group balance in the hypergraph generally helps mitigate prediction bias, which is consistent with our theoretical analysis. Additionally, we observe that an overly rebalanced structure may slightly impair model utility. For instance, on the Bail dataset, as β increases from 0 to 0.5, the model utility improves. However, when β exceeds 0.5, the utility begins to decline. Meanwhile, on the Credit dataset, as the balance level increases, the model almost experiences a consistent decline in predictive performance, with accuracy dropping by about 0.4% overall. This is likely due to large structural perturbations that lead to information loss. Therefore, selecting an appropriate β is crucial for achieving a better trade-off between predictive capability and fairness.

APPENDIX E
RELATED WORKS ON HYPERGRAPH CONTRASTIVE LEARNING

Hypergraph contrastive learning has received little attention and remains largely unexplored. While some investigations in recommender systems have applied contrastive strategy in hypergraphs [11], [12], [15], they are not designed for general



(a) German (b) Bail (c) Credit

Fig. A2: Parameter sensitivity analysis on the rebalance coefficient β. The left y-axis represents the utility metric (ACC), while the right y-axis corresponds to the fairness metrics (Δ_{DP}, Δ_{EO}).

HCL. HyperGCL [9] is the first general HCL framework. It designs a variational auto-encoder to generate augmentation views that can preserve the high-order information in the original hypergraph. Then, HyperGCL-AdT [13] introduces a dual-level contrast mechanism to capture both individual and group-wise behaviors simultaneously, along with an adaptive temperature optimization strategy. TriCL [14] utilizes tri-directional contrastive objectives to further capture group-level and membership-level information in hypergraphs. However, existing methods select a large number of negative samples arbitrarily, which can be both unreliable and inefficient.